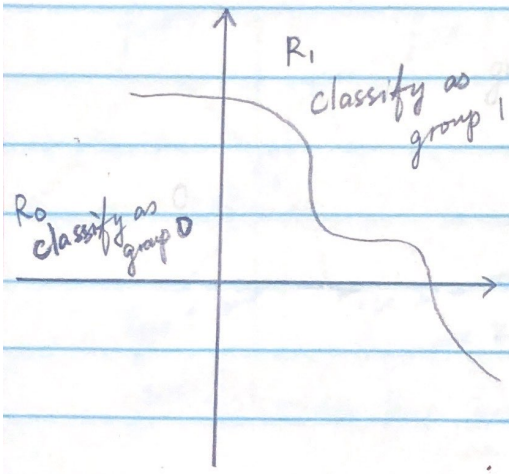
Another derivation of Bayes optimal classification rule



Goal - Find a boundary in $\vec{X}$ space that minimizes misclassification.

$$\text{Error 1} = P(\text{classify as group 0 but true } Y=1)$$
$$= P(\text{classify as group 0} \mid Y=1)\, P(Y=1)$$
$$= P(X \in R_0 \mid Y=1)\, \pi_1$$
$$= \pi_1 \int_{R_0} f_1(x)\, dx \qquad f(x_1) = p(x \mid Y=1)$$

$$\text{Error 2} = P(X \in R_1 \mid Y=0)\, \pi_0 = \pi_0 \int_{R_1} f_0(x)\, dx$$

$$\text{Error 1} + \text{Error 2} = \pi_1 \int_{R_0} f_1(x)\, dx + \pi_0 \int_{R_1} f_0(x)\, dx$$
$$= \pi_1 \left(1 - \int_{R_1} f_1(x)\right) + \pi_0 \int_{R_1} f_0(x)\, dx$$
$$= \pi_1 + \int_{R_1} [\pi_0 f_0(x) - \pi_1 f_1(x)]\, dx$$

To $\min(\text{Error 1} + \text{Error 2}) \iff$ choose a region R_1

$$\pi_0 f_0(x) - \pi_1 f_1(x) < 0$$

$$\iff \frac{\pi_0 f_0(x)}{\pi_0 f_0(x) + \pi_1 f_1(x)} < \frac{\pi_1 f_1(x)}{\pi_0 f_0(x) + \pi_1 f_1(x)}$$

$$\iff p(Y=0 \mid x) < p(Y=1 \mid x)$$